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Graded Assignment: Module 1 Summative Assessment Submitted
Module 1 Summary

Your grade: 86.66%

Your latest: 86.66% • Your highest: 86.66% • To pass you need at least 80%. We keep your highest score.

Next item

Module 1 Summative Assessment

Review Learning Objectives

Instructions

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Due: Oct 20, 11:59 PM IST
Attempts: 0 left (1 attempts every 8 hours)

Submitted
View after due date

Resume

Your grade
To pass you need at least 80%. We keep your highest score.
86.66%

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1. True or False: Bayes' Rule can only be applied when we have equal prior probabilities for all hypotheses. **1 / 1 point**

- ○ True
- ◉ False

Correct
Correct! There are no restrictions on the priors except that they scale to 1.

2. A weather forecast predicts that there is an 80% chance of rain tomorrow. However, historical data shows that the forecast is correct only 75% of the time. What can Bayes' rule help us determine once we find out if it rained? **1 / 1 point**

- ○ The probability of it raining the following day.
- ○ The probability of it not raining.
- ◉ The probability of the forecast being correct.
- ○ None of the above

Correct
Correct! Bayes' rule can help us calculate the updated probability of the forecast being correct using historical data as a prior.

3. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes) and you start with a 99% certainty that it is fair. What is the minimum number of heads in a row you need to flip for your posterior probability that the coin is fair to drop below 15%? **1 / 1 point**

- ◉ 14
- ○ 8
- ○ 18
- ○ It is impossible to determine

Correct
Correct! You need to flip a minimum of 14 heads in a row for your posterior probability that the coin is fair to drop below 15%.

4. Suppose the frequency of a disease in the population is 0.01. The true positive rate of a test is 95% and the true negative rate is 90% (meaning false positives are 10%). What is the fewest number of positive test results that will yield a posterior probability of having the disease higher than 95%? **1 / 1 point**

- ◉ 4
- ○ 1
- ○ 3
- ○ 2

Correct
Four yields a posterior probability of over 95%

5. Suppose you are attempting to determine which two distributions best describe a population. The distribution A is a uniform distribution between 0 and 1. The distribution B is a triangular distribution between 0 and 1 (a height of 0 at 0 and a height of 2 at 1). You have no reason to believe one is more likely than another. Which of the following would increase the posterior probability that the distribution should be A? **1 / 1 point**

- ○ y = .75
- ○ (.35, .65)
- ◉ (0 .25)
- ○ (.5, 1)

Correct
Correct! Under distribution A, this sample has a probability of .25. Under distribution B, this sample has a probability of 0.625. So it is more likely to happen with distribution A.

6. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes). You believe the chance of it being fair is the same as if it is not. You perform an experiment and flip the coin 3 times, all of which turn up tails. Using Bayes' rule, what would you estimate to be the probability that the coin is fair given the data you have collected? **1 / 1 point**

- ◉ 33.3%
- ○ 50%
- ○ 46%
- ○ 13%

Correct
Correct! The posterior probability ends up being 1/3.

7. Suppose the frequency of a disease in the population is 0.01. The true positive rate of a test is 95% and the true negative rate is 90% (meaning the false negative rate is 10%). What is the posterior probability of having the disease given a positive test result? **1 / 1 point**

- ◉ 0.088
- ○ 0.0095
- ○ 0.95
- ○ 0.999

Correct
Correct! We want to know the probability of having the disease given a positive test. The probability of a positive result given that you have the disease is 95% but the disease only occurs in 1% of the population. Our calculation should be (0.95 x 0.01) / ((0.95 x 0.01) + (0.10 x 0.99))

8. A court case is considering if an ICU nurse is guilty of murdering 4 of her patients. She was the only nurse who served all four patients. The prosecution argues that the likelihood of the 4 patients dying, assuming no foul play, is only 0.5%. Based on this, the jury deems her guilty. Were they correct to conclude this? **1 / 1 point**

- ○ No because the likelihood isn't low enough
- ○ Yes because the likelihood is low
- ◉ No because this is the wrong probability to consider
- ○ Yes because she was the only one on duty and the likelihood of her innocence is 0.5%

Correct
Correct! The jurors are supposed to evaluate the probability that the nurse is innocent given the data. What they are given is the probability of the data given that the nurse is innocent. These are not the same at all.

9. Suppose the frequency of a disease in the population is 0.01. The true positive rate of a test is 95% and the true negative rate is 90% (meaning the false positive is 10%). What is the posterior probability of having the disease given TWO positive test results? **1 / 1 point**

- ○ 90%
- ○ 22%
- ◉ 48%
- ○ 16%

Correct
Correct! The probability of having the disease rises significantly with two positive tests.

10. Suppose a pregnancy test has a false positive rate of 1% and a true positive rate of 99%. Also suppose an individual regularly uses contraception that is 99% effective every time they have sexual intercourse. Presume this to be your prior. If after a full cycle with regular sexual activity, this individual receives a positive pregnancy test, what is the posterior probability of pregnancy? **0 / 1 point**

- ○ 75%
- ○ 50%
- ○ There is not enough information to compute this
- ◉ 99%

Incorrect
While the true positive rate of the pregnancy test is 99%, this doesn't take into account the prior probability of pregnancy, which is affected by the individual's regular use of contraception.

11. The posterior will improve with more, randomly selected points of data. **0 / 1 point**

- ○ True
- ◉ False

Incorrect
Bayes' rule allows for updates in the face of new information. A large, unbiased sample will better reflect related and yield better results.

12. True or False: The likelihood is the probability of the data under the hypothesis. **1 / 1 point**

- ◉ True
- ○ False

Correct
Correct!

13. Suppose you aim to spend less time on social media. You install an app that promises to reduce your social media time by 50%. After the first week of using it, your total time spent on social media went down by 12%. What is your posterior probability about the app? **1 / 1 point**

- ○ 61%
- ◉ There is not enough information
- ○ 50%
- ○ 73%

Correct
Correct! You do not know, for example, the standard deviation of your usage. Is a 19% drop between two weeks part of your normal fluctuation?

14. Suppose you're trying to determine if a die is fair (meaning each face is equally likely), and you start with a 99% certainty that it's fair. What is the minimum number of times you need to roll a six in a row for your posterior probability that the die is fair to drop below 10%? **1 / 1 point**

- ◉ 5
- ○ 7
- ○ There is insufficient information
- ○ 6

Correct
Correct! Considering all possible probabilities p for rolling a 6, we see that 5 of them in a row is so unlikely that our posterior dips below 10% despite starting with 99% certainty.

15. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes). You perform an experiment and flip the coin 4 times, all of which turn up tails. Suppose you believe there is a 99% chance the coin is fair, but your friend is more skeptical and begins with a prior of 50%. Which of you will see the biggest difference between your prior and your posterior? **1 / 1 point**

- ◉ Your friend
- ○ You
- ○ You are tied
- ○ It is impossible to tell

Correct
Correct! Your friend starts with a more neutral prior (50%), meaning the data (4 tails) will have a larger impact on changing their posterior probability. This results in a bigger difference between their prior and posterior compared to yours.